

Task – 7

Job Market & Skill Demand Analytics – 20 Feb 2026

Tool Used: Power BI

Dataset Source: Kaggle

Task description:

This project analyzes job market trends using job listings data collected from job portals and LinkedIn datasets. The objective was to identify:

- Most demanded skills
- Salary trends
- City-wise hiring demand
- Emerging technologies
- Work type distribution

The dashboard provides a data-driven view of employment trends and hiring patterns across skills, cities, and job types.

Dataset Overview

The dataset consists of aggregated job postings from multiple sources including job portals and LinkedIn.

Key Fields Used:

- Job ID
- Job Title
- Skill
- City / Location
- Salary (where available – 33% records)
- Work Type (Full-time, Contract, Internship, etc.)
- Posted Date
- Remote Indicator

Dataset Size:

- **Total Job Postings:** 99.05K
- **Remote Opportunities:** 15%
- Salary data available for 33% of postings

The dataset was cleaned by:

- Removing duplicates
- Standardizing city and skill names
- Converting salary into consistent format
- Handling missing salary values

Data Cleaning Steps

1. Removal of Duplicate Records

- Identified duplicates using Job ID + Job Title + Company + City
- Removed repeated listings to avoid inflated job counts
- Ensured each posting was counted only once

Impact: Improved accuracy of total job postings (99.05K).

2. Handling Missing Values

Salary Data

- Salary available only for 33% of records
- Missing salary fields were left blank (not replaced with 0)
- Salary analysis performed only on available records

Impact: Prevented distortion of median salary calculations.

Skill Data

- Cleaned empty/null skill entries
- Removed irrelevant symbols or formatting issues

3. Standardization of City Names

Observed inconsistencies such as:

- “NYC” vs “New York”
- “Bangalore” vs “Bengaluru”

Standardized city names to maintain consistency in city-wise analysis.

Impact: Accurate Top Hiring Cities and Salary by City visuals.

4. Skill Normalization

Skills were originally stored as comma-separated text.

Processing included:

- Splitting skills into individual rows
- Removing extra spaces
- Converting all skills to uppercase for consistency
- Standardizing variations (e.g., “SQL Server” grouped under SQL where appropriate)

Impact: Enabled accurate calculation of:

- Most Demanded Skills
- Emerging Skills
- Salary by Skill

5. Salary Processing

- Converted salary ranges into a median value (where applicable)
- Standardized salary format to annual USD equivalent
- Removed unrealistic outliers (extremely high/low salary values)

Impact: Reliable Salary Trend and Salary by Skill analysis.

6. Date Formatting & Trend Preparation

- Converted posted date to proper Date format
- Extracted Year and Month columns
- Created Month-Year hierarchy

Impact: Enabled Job Posting Trend (2023–2024) visualization.

7. Work Type Categorization

Standardized work types into:

- Full-time
- Contract
- Part-time
- Internship
- Temporary

Impact: Clear Work Type Distribution analysis.

8. Data Transformation (Modeling Preparation)

- Created relationships between Job, Skill, and Location tables
- Built calculated measures:
 - Total Job Count
 - Median Salary
 - Growth %
 - Remote %
- Filtered invalid records before model loading

9. Data Validation

Final checks included:

- Cross-verification of total job count
- Recalculation of salary median
- Checking skill frequency consistency
- Validating growth percentages

Summary of Data Processing

The data cleaning and transformation process ensured:

- High data accuracy
- Reliable KPI calculations
- Correct skill frequency counts
- Accurate salary analysis
- Meaningful city-wise comparisons

This structured preprocessing approach strengthened the overall credibility and analytical depth of the dashboard.

Dashboard KPIs

Total Job Postings: 99.05K

- Indicates strong job demand in the analyzed domain.

Remote Opportunities: 15%

- Shows that hybrid/on-site roles still dominate the market.

Fastest Growing Skill: SQL

- SQL shows the highest growth rate in the last 30 days.

Visual Analysis & Explanation

1. Job Posting Trend (2023–2024) – Line Chart

This chart shows month-wise job growth.

Observation:

- Significant increase in postings from March to April 2024.
- Indicates strong hiring momentum in early 2024.

Purpose:

Helps identify hiring cycles and seasonal demand.

2. Salary Trend – Line Chart

Displays median salary trend based on 33% available data.

Observation:

- Peak median salary around \$135K.
- Slight correction observed after peak.

Purpose:

Tracks compensation movement and hiring competitiveness.

3. Salary by Skill – Horizontal Bar Chart

Top-paying skills:

- Machine Learning (~\$180K)
- Python (~\$150K)
- SQL (~\$130K)
- Tableau

- Power BI

Insight:

Advanced technical skills command higher salaries.

4. Emerging Skills (Last 30 Days Growth) – Growth % Bar Chart

SQL shows extremely high growth (~24,879%).

Excel, Tableau, Power BI, and Python also show strong growth.

Insight:

Core data skills are experiencing sharp demand spikes.

5. Most Demanded Skills – Bar Chart

Top skills by job count:

- SQL
- Python
- Machine Learning
- Tableau
- Power BI

Insight:

SQL remains foundational across roles.

6. Work Type Distribution – Bar Chart

- Full-time dominates the market
- Contract roles follow
- Internship and temporary roles are minimal

Insight:

Organizations prefer long-term hiring over short-term roles.

7. Salary by City – Bar Chart

Highest median salary observed in:

- Westwood, CA
- Nyack, NY
- Springfield, IL

Insight:

Location significantly influences salary structure.

8. Top Hiring Cities – Bar Chart

Highest job demand in:

- United States (aggregated)
- New York, NY
- Chicago, IL
- Houston, TX
- Atlanta, GA

Insight:

Major metropolitan cities dominate hiring demand.

Key Insights & Observations

Based on dashboard analysis:

- The job market shows rapid growth in early 2024.
- SQL is both the most demanded and fastest-growing skill.
- Machine Learning and Python command premium salaries.
- Only 15% of jobs are remote, showing continued preference for hybrid/on-site roles.
- Full-time roles dominate hiring patterns.
- Salary varies significantly by location.
- Data and analytics skills continue to remain high-demand across cities.

Recommendations

For Job Seekers:

- Prioritize SQL + Python as core skill combination
- Add Machine Learning for higher salary potential
- Consider relocation to high-paying cities

For Organizations:

- Benchmark salaries in competitive markets
- Invest in SQL and data engineering talent
- Expand remote opportunities to widen talent pool

For Training Institutions:

- Focus curriculum on SQL, Python, ML, Tableau, Power BI
- Introduce practical data analytics projects

Conclusion

The job market analysis reveals strong demand for core data and analytics skills, particularly SQL and Python. While advanced skills like Machine Learning command higher salaries, foundational database skills remain essential across industries.

Hiring growth in 2024 suggests expanding digital transformation and data-driven decision-making across organizations.

The dashboard provides valuable insights for:

- Job seekers (skill prioritization)
- HR teams (market benchmarking)
- Training institutions (course alignment)
- Business leaders (talent acquisition strategy)